

A Packing of 16 Four-Spaces in $\mathbb{R}^9$

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October 9, 2026

Abstract

We give 16 4-dimensional subspaces of $\mathbb{R}^9$ whose smallest squared chordal distance is at least 2.370046. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 2.360731, obtained by deleting 1 of the 17 four-spaces of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 16 subspaces of dimension 4 in $\mathbb{R}^9$ whose pairwise squared chordal distances $D(a, b) = 4 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 2.370046 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane's *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 9$, $n = 4$. The cell $N = 16$ is blank in Sloane's table, in Sloane's coordinate files and in Cohn's table as checked on October 8, 2026. The best value implied by known packings is 2.360731, obtained by deleting 1 of the 17 four-spaces of the packing found by this search. The Rankin simplex bound for this cell is $4(9 - 4)/9 \cdot 16/15 = 2.370370$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m9n4N16.txt`

in <https://github.com/Ge-limin/math>.

*Found and checked by Claude, a model released by Anthropic, running on the author's laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 63 seconds on one laptop, with about ten searchers running at once.

A Exact value

numerator:

```
237004607178649941516206979434214205719904253600768232482953509890708193
454561561116116132901799476876996490445699546852563237979290492659611645
6439565701260030779393848566045801858949610850009
```

denominator:

```
999999999997721770247310028569058577406070195674215687626584966790205826
683918117896447420727930841951768334706998115656881623033704710428009111
104973870318613156351135401281668224837588543730
```

B Coordinates

16 blocks of 9 rows with 4 integers each; divide by 10^{12} . File: `grassmann/packings/m9n4N16.txt`.

```
203455667676 606847088490 308084217084 184611854840 113743303827 -107913241466 259503150386 543526821392 20793860282 673902931736 -423757163295 -175209664383
118403436626 76984287885 137200448730 220592634204 544856137555 131898112201 -414083548802 505255491020 465021879 326773215646 520020419228 457920953430
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